David Chiu

Curriculum Vitae

Department of Mathematics and Computer Science
University of Puget Sound
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davidtchiu.github.io

Education

2005 – 2010 **PhD, Computer Science and Engineering**
The Ohio State University | Systems Group
"Automatic Service Planning in Cloud/Grid Environments"
Dissertation Advisor: Gagan Agrawal

1998 – 2004 **BS, MS, Computer Science**
Kent State University
Thesis Advisor: Paul S. Wang

Academic Appointments

2022 – **Professor of Computer Science**
Department of Mathematics and Computer Science
University of Puget Sound, Tacoma, Wash.

2017 – 2022 **Associate Professor of Computer Science (with tenure)**
Department of Mathematics and Computer Science
University of Puget Sound, Tacoma, Wash.

2014 – 2017 **Assistant Professor of Computer Science**
Department of Mathematics and Computer Science
University of Puget Sound, Tacoma, Wash.

2010 – 2014 **Assistant Professor of Computer Science**
School of Engineering and Computer Science
Washington State University, Vancouver, Wash.

2005 – 2010 **Graduate Research and Teaching Associate**
Department of Computer Science and Engineering
The Ohio State University, Columbus, Ohio

2002 – 2004 **Graduate Research and Teaching Associate**
Department of Computer Science
Kent State University, Kent, Ohio

Administrative Appointments

2019 – 2022 **Chair of Computer Science**
Department of Mathematics and Computer Science
University of Puget Sound, Tacoma, Wash.

Teaching Experience (Instructor of Record)

2014 – **University of Puget Sound**

Introduction to Computer Science; Data Structures; Software Engineering; Principles of Database Systems; Operating Systems; Capstone Planning Seminar; Capstone in Computer Science

Summer 2015 **Tunghai University (as Visiting Professor)**

An Introduction to Computer Science

2010 – 2014 **Washington State University, Vancouver**

Introduction to Database Systems; Operating Systems; Web Data Management; Advanced Topics: High-Performance Computing (Graduate)

2005 – 2010 **The Ohio State University (as Graduate Teaching Associate)**

Elementary Programming; Introduction to Programming and Algorithms for Scientists and Engineers

2003 – 2004 **Kent State University (as Graduate Teaching Associate)**

Computer Literacy; Introduction to Computer Science

Publications

Note: My research advisees and I are underlined.

Conference and Journal Articles (Acceptance rates listed if known)

BDCAT 2026 A. Kirner, J. Sawin, and D. Chiu. Dynamic Bitmap Join Execution with Integrated Range Query Support. Proceedings of the 13th IEEE/ACM International Conference on Big Data Computing, Application, and Technologies (BDCAT'26). Florianopolis, Brazil. 2026. **(64 submitted, 17 accepted as full papers, 27% acceptance rate).**

BDCAT 2023 J. Kaeppel, J. Sawin, and D. Chiu. Workload Aware Cache Management of Bitmap Indices. Proceedings of the 10th IEEE/ACM International Conference on Big Data Computing, Application, and Technologies (BDCAT'23). Taormina, Italy. 2023. **Best Paper Award**

SSDBM 2021 S. McClain, M. Mutschler-Aldine, C. Monaghan, D. Chiu, J. Sawin, and P. Jarvis. Caching Support for Range Query Processing on Bitmap Indices. 33rd International Conference on Scientific and Statistical Database Management (SSDBM). Tampa, FL. 2021. ACM. **(59 submitted, 16 accepted, 27.1% acceptance rate).**

DSE 2021 B. Tran, B. Schaffner, J. Myre, J. Sawin, and D. Chiu. Exploring Means to Enhance the Efficiency of GPU Bitmap Index Query Processing. Data Science and Engineering. Vol 6(2). Springer. 2021.

JCC 2020 M. Nelson, Z. Sorenson, J. Myre, J. Sawin, and D. Chiu. Parallel Acceleration of CPU and GPU Range Queries over Large Data Sets. Journal of Cloud Computing. Vol 9(44). Springer. 2020.

DASFAA 2020 B. Tran, B. Schaffner, J. Sawin, J. Myre, and D. Chiu. Increasing the Efficiency of GPU Bitmap Index Query Processing. 25th International Conference on Database Systems for Advanced Applications (DASFAA'20). JeJu, South Korea. 2020. **(487 submitted, 121 accepted, 25% acceptance rate).**

BDCAT 2019 M. Nelson, Z. Sorenson, J. Myre, J. Sawin, and D. Chiu. GPU Acceleration of Range Queries over Large Data Sets. Proceedings of the 6th IEEE/ACM International Conference on Big Data Computing, Application, and Technologies (BDCAT'19). Auckland, New Zealand. 2019. **(47 submitted, 13 accepted, 27.7% acceptance rate). Best Paper Finalist**

CLOUD 2019 J. Polonitza, D. Chiu, and B. Ren. A Transactional Framework for Broadening Access to Geo-Diversification. Proceedings of the 12th IEEE International Conference on Cloud Computing (Cloud'19). Milan, Italy. 2019. **(139 submitted, 29 accepted as short papers, 26% short-paper acceptance rate)**

BDCAT 2018 S. Burdick, J. Risner, D. Chiu, and J. Sawin. Fault-Tolerant Query Execution over Distributed Bitmap Indices. Proceedings of the 5th IEEE/ACM International Conference on Big Data Computing, Application, and Technologies (BDCAT'18). Zurich, Switzerland. 2018 **(101 submitted, 23 accepted, 22.8% acceptance rate). Best Paper Finalist**

PES 2018 R. Bass, J. Landford, R. Meier, B. McCamish, E. Cotilla-Sanchez, and D. Chiu. Event Detection Using Correlation within Arrays of Streaming PMU Data. Proceedings of the 2018 IEEE Power and Energy Society General Meetings (PES'18). Portland, OR. 2018.

IDEAS 2017 B. Taufen, J. Sawin, and D. Chiu. Improving the Querying Efficiency of the PLWAH Bitmap Algorithm. 21st International Database Applications and Engineering Symposium (IDEAS'17). Bristol, UK. 2017 **(30% acceptance rate)**

CLOUD 2016 C. Johnson and D. Chiu. Hadoop in Flight: Migrating Live MapReduce Jobs for Power-Shifting Data Centers. 9th IEEE International Conference on Cloud Computing (Cloud'16). San Francisco, CA. 2016 **(327 submitted, 49 accepted, 15% research-track acceptance rate)**

IDEAS 2016 G. Guzun, G. Canahuate, and D. Chiu. A Two-Phase MapReduce Algorithm for Scalable Preference Queries over High-Dimensional Data. In Proceedings of the 20th International Database Engineering and Applications Symposium (IDEAS'16). Montreal, Canada. 2016 **(127 submitted, 19 accepted as full papers, 15% acceptance rate)**

FICLOUD 2016 M. Velez, J. Sawin, A. Ingerson, and D. Chiu. Improving Bitmap Execution Performance Using Column-Based Metadata. 4th IEEE International Conference on Future Internet of Things and Cloud (FiCloud'16). Vienna, Austria. 2016 **(30% acceptance rate)**

EPSR 2016 B. McCamish, R. Meier, J. Landford, R. Bass, D. Chiu, and E. Cotilla-Sanchez. A Backend Framework for the Efficient Management of Power System Measurements. Electric Power Systems Research. 2016.

CSN 2016 E. Otsuka, S. Wallace, and D. Chiu. A Hashtag Recommendation System for Twitter Data Streams. Computational Social Networks. Vol 3(3). Springer. 2016.

CCBD 2015 X. Xu, X. Zhao, G. Dunham, D. Chiu, and J. Xu. Modeling Parallel Simulations over Amazon EC2. International Conference on Cloud Computing and Big Data (CCBD'15), Taipei, Taiwan. 2015. **(50% short-paper acceptance rate)**

PACLIC 2015 X. Huang, T. Liu, D. Chiu, X. Li, and T. Zhu. Topic Model for Identifying Suicidal Ideation in Chinese Microblog. 29th Pacific Asia Conference on Language, Information and Computation (PACLIC'15). Shanghai, China. 2015 **(28% acceptance rate)**

ICPADS 2014 B. Wang, X. Zhao, and D. Chiu. Lightweight Online Power Monitoring and Control for Mobile Applications. In Proceedings of the 20th IEEE International Conference on Parallel and Distributed Systems (ICPADS'14), Hsinchu, Taiwan. 2014 **(322 submitted, 97 accepted, 30% acceptance rate)**

IDEAS 2014 R. Slechta, J. Sawin, B. McCamish, D. Chiu, and G. Canahuate. Optimizing Query Execution for Variable-Aligned Length Compression of Bitmap Indices. In Proceedings of the 18th International Database Engineering and Applications Symposium (IDEAS'14), Porto, Portugal. 2014 **(19.7% full-paper acceptance rate)**

IDEAS 2014 E. Otsuka, S. Wallace, and D. Chiu. Design and Evaluation of a Twitter Hashtag Recommendation System. In Proceedings of the 18th International Database Engineering and Applications Symposium (IDEAS'14). Porto, Portugal. 2014. **(short paper; 36% short-paper acceptance rate)**

ICDE 2014 G. Guzun, G. Canahuate, D. Chiu, and J. Sawin. A Tunable Aligned Compression Framework for Bitmap Indices. In Proceedings of the 30th International Conference on Data Engineering (ICDE'14). Chicago, IL. 2014 **(446 submitted, 89 accepted, 19.9% acceptance rate)**

UICW 2014 X. Huang, L. Zhang, D. Chiu, X. Li, and T. Zhu. Detecting Suicide Ideation in Chinese Microblog with Psychological Lexicons. In Proceedings of the 2014 Workshop on Pervasive and Ubiquitous Data Analytics. Held in conjunction with the 11th IEEE International Conference on Ubiquitous Intelligence and Computing.

SusTech 2014 R. Meier, M. Histand, J. Landford, B. McCamish, D. Chiu, R. Bass, and E. Cotilla-Sanchez. Power System Data Management and Analysis Using Synchrophasor Data. In Proceedings of the 2nd IEEE Conference on Technologies for Sustainability (SusTech'14). Portland, OR. 2014

SusTech 2014 B. McCamish, D. Chiu, M. Histand, R. Meier, J. Landford, E. Cotilla-Sanchez, and R. Bass. Managing PMU Data Sets with Bitmap Indexes. In Proceedings of the 2nd IEEE Conference on Technologies for Sustainability (SusTech'14). Portland, OR. 2014

IJNGC 2014 F. Kabir, T. Hall, S. Wallace, and D. Chiu. Elastic Resource Allocation for a Cloud-Based Web Caching System. International Journal of Next-Generation Computing. Vol 5(1). 2014.

JCSC 2013 D. Chiu and S. Wallace. On the Science in Computer Science: Integrating Research Preparedness in Undergraduate CS. Journal of Computing Sciences in Colleges (CCSC-NW). Vol 29(1). 2013.

TSC 2013 D. Chiu and G. Agrawal. Cost and Accuracy Aware Scientific Workflow Composition for Service-Oriented Environments. IEEE Transactions on Services Computing. Vol 6(4). IEEE. 2013.

IDEAS 2013 F. Doan, D. Chiu, B. Perez Lukes, J. Sawin, G. Guzun, and G. Canahuate. Dynamic Bitmap Recompression through Workload-Based Optimizations. In Proceedings of the 17th International Database Engineering and Applications Symposium (IDEAS'13). Barcelona, Spain. 2013 **(113 submitted, 14 accepted 12.4% full paper acceptance rate)**

DMC 2013 Y. Li, D. Chiu, C. Liu, L. T. X. Phan, T. Gill, S. Aggarwal, Z. Zhang, B. T. Loo, D. Maier, and B. McManus. Towards Dynamic Pricing-Based Collaborative Optimizations for Green Data Centers. In Proceedings of the 2nd International Workshop on Data Management in the Cloud (DMC'13). Workshop Proceedings of ICDE. Brisbane, Australia.

IPDPS 2013 T. Biçer, J. Yin, D. Chiu, G. Agrawal, and K. Schuchardt. A Compression System for Accelerating Large-Scale Data Analytics Applications. In Proceedings of the 27th IEEE International Parallel and Distributed Processing Symposium (IPDPS'13). Boston, MA. 2013 **(494 submitted, 106 accepted, 21% acceptance rate)**

CSC 2012 F. Kabir and D. Chiu. Reconciling Cost and Performance Objectives for Elastic Web Caches. In Proceedings of the 2012 IEEE International Conference on Cloud and Services Computing (CSC'12). Shanghai, China. 2012 **(110 submitted, 36 accepted, 33% acceptance rate)**

CCGRID 2012 T. Biçer, D. Chiu, and G. Agrawal. Time and Cost Sensitive Data-Intensive Computing on Hybrid Clouds. In Proceedings of the 2012 IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid'12). Ottawa, Canada. 2012 **(302 submitted, 83 accepted, 27% acceptance rate)**

PER 2012 D. Chiu, C. Stewart, and B. McManus. Electric Grid Balancing through Low-Cost Workload Migration. ACM SIGMETRICS Performance Evaluation Review. Vol 40(3). ACM. 2012.

CPE 2012 V. T. Ravi, W. Ma, D. Chiu, and G. Agrawal. Compiler and Runtime Support for Enabling Reduction Computations on Heterogeneous Systems. Concurrency and Computation: Practice and Experience. Vol 24(5). Wiley. April, 2012.

IJNGC 2011 D. Chiu, T. Hall, F. Kabir, A. Shetty, and G. Agrawal. Analyzing Costs and Optimizations for an Elastic Key-Value Store on Amazon Web Services. International Journal on Next-Generation Computing Vol. 2(2). Special Issue on Cloud Computing. July 2011.

DEXA 2011 F. Corrales, D. Chiu, and J. Sawin. Variable Length Compression for Bitmap Indices. In Proceedings of the 22nd International Conference on Database and Expert Systems Applications (DEXA'11). Toulouse, France. 2011 **(356 submitted, 89 accepted, 25% acceptance rate)**

IDEAS 2011 D. Chiu, T. Hall, F. Kabir, and G. Agrawal. An Approach towards Automatic Workflow Composition through Information Retrieval. In Proceedings of the 15th International Database Engineering and Applications Symposium (IDEAS'11). Lisbon, Portugal. 2011 **(130 submitted, 20 accepted as full papers, 15.4% acceptance rate)**

CLUSTER 2011 T. Biçer, D. Chiu, and G. Agrawal. A Framework for Data-Intensive Computing with Cloud Bursting. In Proceedings of the 2011 IEEE International Conference on Cluster Computing (Cluster'11). Austin, TX. 2011 **(28% acceptance rate)**

CCGRID 2011 D. Chiu, A. Shetty, and G. Agrawal. Evaluating and Optimizing Indexing Schemes for a Cloud-based Elastic Key-Value Store. In Proceedings of the 11th IEEE/ACM Symposium on Cluster, Cloud, and Grid Computing (CCGrid'11). Newport Beach, CA. 2011. **(189 submitted, 55 accepted, 29% acceptance rate)**

MTAGS 2011 T. Biçer, D. Chiu, and G. Agrawal. MATE-EC2: A Middleware for Processing Data with AWS. 4th International Workshop on Many-Task Computing on Grids and Supercomputers (MTAGS'11). Seattle, WA. 2011 **Invited**.

SC 2010 D. Chiu, A. Shetty, and G. Agrawal. Elastic Cloud Caches for Accelerating Service-Oriented Computations. In Proceedings of the 23rd ACM/IEEE International Conference on High Performance Computing Networking, Storage and Analysis (SC'10). New Orleans, LA. 2010. **(253 submitted, 51 accepted, 20% acceptance rate)**

GRID 2010 D. Chiu and G. Agrawal. Evaluating Caching and Storage Options on the Amazon Web Service Cloud. In Proceedings of the 11th ACM/IEEE International Conference on Grid Computing (GRID'10). Brussels, Belgium. 2010. **(124 submitted, 28 accepted, 22.6% acceptance rate)**

ICS 2010 V. T. Ravi, W. Ma, D. Chiu, and G. Agrawal. Compiler and Runtime Support for Enabling Generalized Reduction Computations on Heterogeneous Parallel Configurations. In Proceedings of the 24th ACM/SIGARCH International Conference on Supercomputing (ICS'10). Tsukuba, Japan. 2010. **(180 submitted, 32 accepted, 17.8% acceptance rate)**

XRDS 2010 D. Chiu. Profile Hiroshi Ishii: Tangible Bits. ACM Crossroads. 16(4), 2010.

XRDS 2010 D. Chiu. Elasticity in the Cloud. ACM Crossroads. 16(3), 2010.

ICWS 2009 D. Chiu, S. Deshpande, G. Agrawal, and R. Li. A Dynamic Approach toward QoS-Aware Service Workflow Composition. In Proceedings of the 7th IEEE International Conference on Web Services (ICWS'09). Los Angeles, CA. 2009. **(18% acceptance rate)**

SSDBM 2009 D. Chiu and G. Agrawal. Enabling Ad Hoc Queries over Low-Level Scientific Data Sets. In Proceedings of the 21st International Conference on Scientific and Statistical Database Management (SSDBM'09). New Orleans, LA. 2009. **(76 submitted, 29 accepted, 38% acceptance rate)**

CCGRID 2009 D. Chiu and G. Agrawal. Hierarchical Caches for Grid Workflows. In Proceedings of the 9th IEEE/ACM Symposium on Cluster Computing and the Grid (CCGrid'09). Shanghai, China. 2009. **(271 submitted, 57 accepted, 21% acceptance rate)**

GRID 2008 D. Chiu, S. Deshpande, G. Agrawal, and R. Li. Cost and Accuracy Sensitive Dynamic Workflow Composition over Grid Environments. In Proceedings of the 9th IEEE/ACM International Conference on Grid Computing (GRID'08). Tsukuba, Japan. 2008. **(176 submitted, 35 accepted, 19.8% acceptance rate)**

XRDS 2008 D. Chiu. On Teaching Computer Science: Thoughts and Advice for TAs. ACM Crossroads. 15(2), 2008.

DSMM 2007 F. Altiparmak, D. Chiu, and H. Ferhatosmanoglu. Incremental Quantization for Aging Data Streams. ICDM Workshop on Data Stream Mining and Management (DSMM'07). Omaha, NE. 2007

ITCC 2004 D. Chiu. Web-based Mathematics Education with MathChat. In Proceedings of the IEEE International Conference on Information Technology: Coding and Computing (ITCC'04). Las Vegas, NV. 2004. **Best Student Paper Award**.

IAMC 2003 D. Chiu, Y. Zhou, X. Zhu, and P. S. Wang. Design, Implementation, and Processing Support of MeML. ACM ISSAC Workshop on Internet Accessible Mathematics Computation (IAMC'03). Philadelphia, PA. 2003.

Posters and Poster Papers

CCSC 2018 J. Polonitza and D. Chiu. Towards a Transactive Future. 20th Consortium for Computing Sciences in Colleges (CCSC-NW 2018). **Best Poster Runner-up**.

CCSC 2018 S. Walling-Bell and D. Chiu. EEG Experiment Scripting Tool for Novice Programmers. 20th Consortium for Computing Sciences in Colleges (CCSC-NW 2018).

ICAC 2017 A. Ingerson, D. Chiu, and J. Sawin. Cache-Friendly Bitmap Compression on Symmetric Multiprocessors. 14th IEEE International Conference on Autonomic Computing (ICAC 2017). Columbus, Ohio.

CCSC 2016 R. Hirsch and D. Chiu. Live Data Compression, Caching, and Querying of Bitmap Indices. 18th Consortium for Computing Sciences in Colleges (CCSC-NW 2016)

CCSC 2015 A. Ingerson and D. Chiu. Improving the Performance of Parallelized Bitmap Index Compression through Data Striping. 17th Consortium for Computing Sciences in Colleges (CCSC-NW 2015). **Best Poster Award**.

SPLASH 2015 B. McCamish, X. Zhao, D. Chiu, J. Sawin, and G. Canahuate. Evaluating Work Distribution Patterns for Parallel Bitmap Compression for SMPs. ACM SIGPLAN conference on Systems, Programming, Languages and Applications: Software for Humanity (SPLASH'15)

MobiSys 2014 B. Wang, X. Zhao, and D. Chiu. A Power-Aware Mobile App for Field Scientists. 12th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys'14)

EDUPAR 2014 X. Zhao, S. Wallace, and D. Chiu. Integrating PDC Topics in Multiple Levels of CS Courses at WSU Vancouver. NSF/TCPP Workshop on Parallel and Distributed Computing Education (EduPar'14), in conjunction with IPDPS'14

WSUV 2013 G. Dunham, D. Chiu, X. Zhao, and J. Xu. Resource Allocation for Molecular Simulations Using Cloud Computations. WSU Vancouver Research Showcase 2013. **Best Undergraduate Poster Award.**

SSDBM 2011 D. Chiu, T. Hall, F. Kabir, and G. Agrawal. Keyword Search Support for Automating Scientific Workflow Composition. In Proceedings of the 23rd International Conference on Scientific and Statistical Database Management (SSDBM'11). Portland, OR. 2011

SC 2011 T. Biçer, D. Chiu, and G. Agrawal. A Framework for Data-Intensive Computing with Cloud Bursting. 24th ACM/IEEE International Conference on High Performance Computing Networking, Storage and Analysis. Seattle, WA. 2011

SC,GHC 2011 F. Kabir and D. Chiu. A Self-Managed Cloud Cache for Accelerating Data-Intensive Applications. 24th ACM/IEEE International Conference on High Performance Computing Networking, Storage and Analysis; Also, Grace Hopper Celebration

GIS 2008 D. Chiu, S. Deshpande, G. Agrawal, and R. Li. Composing Geoinformatics Workflows with User Preferences. In Proceedings of the 16th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS'08). Irvine, CA. 2008

eScience 2007 D. Chiu and G. Agrawal. Ad Hoc Scientific Workflows through Data-driven Service Composition. 2007 Microsoft eScience Workshop. RENCI, Chapel Hill, NC. 2007

Book Chapters

2010 D. Chiu and G. Agrawal. Auspice: Automatic Service Planning in Cloud/Grid Environments. Book Chapter in Grids, Clouds, and Virtualization. Massimo Cafaro and Giovanni Aloisio (Eds.), Springer-Verlag. 2010

PhD Dissertation

2010 D. Chiu. AUSPICE: Automatic Service Planning in Cloud/Grid Environments. Doctoral Thesis. The Ohio State University. 196 pages. June, 2010. Committee: Gagan Agrawal, Christopher Stewart, Feng Qin, and Hakan Ferhatosmanoglu.

GRANTS

External Funding

2026 **NSF XSEDE Allocation Grant (PI)**, "Education, Research, and Training of Cloud Resources at Puget Sound". Multiple awards in: 2026, 2022, 2021, 2020, 2019. 200,000 SUs (≈ $3,200 per award).

2022 **Student Support for IEEE/ACM UCC 2022 and BDCAT 2022 (co-PI)**, National Science Foundation ($32,000)

2017 **Faculty Travel Grant**, IEEE International Conference on Autonomic Computing

2015 **GooeySoft Analytics Grant (PI)** ($7,954)

2013 **Schweitzer Engineering Labs** SyncrhoWAVE Software Donation ($5,000)

2013 **Oregon BEST Center (PI)**: "BPA Special Study on Synchrophasors: Big Data" with R. Bass and E. Cotilla-Sanchez ($69,504)

2013 **NSF TPCC Early Adopter Award (PI)**, with X. Zhao and S. Wallace. ($2,500)

2012 **Amazon Education Grant (PI)**: "Web Data Management" in years 2012 and 2011 ($4,500 + $2,100)

2011 **M.J. Murdock Charitable Trust Grant (co-PI)**: "Interdisciplinary Renewable Energy Option Track" with W. Xue, S. Solivitz, and H. Gurocak ($250,490)

2011 **Broader Engagement and Mentoring Grant**: SC'12. (∼$2,000)

2010 **Amazon Research Grant (PI)**: "Cloud-Based Strategies for Accelerating Scientific Computing" ($7,500)

2010 **Broader Engagement and Mentoring Grant**: SC'11. (∼$2,000)

Internal Funding

2026 **John Lantz Senior Fellowship for Research or Advanced Study**, University of Puget Sound. (Full year sabbatical + $4,500 travel award)

2026 **McCormick Research Grant**, University of Puget Sound. Multiple awards in years: 2026; 2023; 2015 ($4,000 each)

2023 **UEC Faculty Travel Grant**, University of Puget Sound. Multiple awards in years: 2023; 2018; 2017; 2016; 2014. (Various amounts)

2020 **Agricola Faculty Research Award**, University of Puget Sound ($1,000)

2020 **Burlington Northern Curriculum Development Grant**. "Data Science Major." University of Puget Sound ($6,540)

2020 **Lind-Van Enkevort (LvE) Award**, with A. Chambers. "Software Engineering curriculum." University of Puget Sound ($6,000)

2017 **Lind-Van Enkevort (LvE) Award**, University of Puget Sound ($4,500)

2016 **Lind-Van Enkevort (LvE) Award**, University of Puget Sound ($6,000)

2011 **WSU New Faculty Seed Research Grant**: "Accelerating Large-Scale Workflow Processes over a Cloud Environment" ($17,057)

2011 **WSU Vancouver Mini Research Grant**: "A Self-Tuning Bitmap Compression Framework for Fast Data Processing" ($2,770)

Service to the University

Leadership (Elected or Appointed)

2023 - 2027 **Director and Advisor**, Mosaic Residential Community, University of Puget Sound (appointed by VP of Student Affairs)

2025 - 2026 **Chair**, Academic Standards Committee (ASC), University of Puget Sound

2024 - 2025 **Co-chair**, International Education Committee (IEC), University of Puget Sound

2023 - 2025 **Member**, Total Compensation and Benefits (TCB) Project, University of Puget Sound (appointed by the Office of the President)

2021 **Member**, Executive VP and CFO Search Committee, University of Puget Sound (appointed by the Office of the President)

2019 - 2022 **Department Chair**, Department of Mathematics and Computer Science, University of Puget Sound (appointed by the Provost)

2018 - 2020 **Member**, Curriculum Task Force (CTF), University of Puget Sound (elected)

2011 - 2014 **Graduate Studies Chair**, Computer Science, WSU Vancouver (appointed by the Director of School of Engineering)

University Committees

2024 - present **Member**, Data Analytics Program Committee, University of Puget Sound

2025 - 2026 **Member**, Academic Standards Committee (ASC), University of Puget Sound

2024 - 2025 **Member**, International Education Committee (IEC), University of Puget Sound

2021 - 2024 **Member**, Student Life Committee (SLC), University of Puget Sound

2020 - 2021 **Member**, Lantz Sabbatical Awards Committee, University of Puget Sound

2020 - 2021 **Member**, High Impact Practices Working Group, (Continued work of the CTF — Established the *"Experiential Learning"* graduation requirement)

2015 - 2018 **Member**, Curriculum Committee (CC), University of Puget Sound

2017 - 2018 **Member**, Alumni Mentoring Exploratory Committee, University of Puget Sound

2017 - 2018 **Member**, Data Analytics Faculty Workgroup, University of Puget Sound

2016 - 2017 **Alternate Member**, Faculty Senate, University of Puget Sound

2011 - 2014 **University Scholars (Honors) Faculty Advisor**, WSU Vancouver

2009 - 2010 **Graduate Steering Committee**, The Ohio State University

2007 - 2008 **Council of Graduate Students**, The Ohio State University

Departmental Committees

2026 - 2027 **Chair, CS Search Committee**, University of Puget Sound

2023 - present **Lind-Van Enkevort (LvE) Award Committee**, University of Puget Sound

2022 - 2023 **Mathematics Search Committee**, University of Puget Sound

2020 - 2021 **Chair, Data Science Major Committee**, University of Puget Sound

2020 - 2021 **Data Science Search Committee (Chair)**, University of Puget Sound

2021 - 2022 **Computer Science Search Committee**, University of Puget Sound

2014 - 2015 **Lind-Van Enkevort (LvE) Award Committee**, University of Puget Sound

2014 - 2015 **CS Search Committee**, University of Puget Sound

Campus Engagement and Outreach

2025 - 2026 **Board Member**, Faculty Development Center (FDC), University of Puget Sound

2024 - 2026 **Panelist**, *"Supporting First-Generation Students"* @ Peer Advisor Training

2025 **Presenter**, ***"Grading and Equity in the Age of AI"*** @ Faculty Teaching Workshop

2025 **Site Evaluator**, DIS Study Abroad Program in Scandinavia

2024 - 2025 **New Faculty Mentor**, Faculty Development Center, University of Puget Sound

2024 - 2025 **Faculty Search Committee**, Communication Studies, University of Puget Sound

2024 - 2025 **Faculty Career Champion**, University of Puget Sound

2024 **Speaker**, *"Bitmap Indexing for the Management of Big Data,"*. Alumni College of the 2024 Summer Reunion, University of Puget Sound

2023 **Participant**, Taiwan Intercultural Engagement, University of Puget Sound

2022 **Speaker**, *"Bitmap Indexing for the Management of Big Data,"*. Visible Spectrum's 7th Annual Research Symposium, University of Puget Sound

2021 **Panelist**, *"I Am Puget Sound"* Student Orientation Program

2021 **Panelist**, CS/Math Graduate School Panel

2020 **Panelist**, Wednesday @ 4, *"Retooling Syllabi Mid-semester to Achieve Learning Objectives [during COVID-19]"*

2020 **Panelist**, CS/Math/Physics Graduate School Panel

2018 **Site Evaluator**, AIT-Budapest Study Abroad Program

2017 **Panelist**, Asian/Pacific Islander Faculty and Staff Panel, APIC

2015 **CS Representative**, Decision Puget Sound

Service to Profession

External Evaluator

2025 **Reviewer, CS bachelors degree program**, Tacoma Community College

2023 **Reviewer, Faculty Promotion**, Washington State University

2020 **Reviewer**, United States Citizenship and Immigration Services (USCIS)

2020 **Reviewer, Faculty Tenure and Promotion**, Rhodes College

Conference Organization

BDCAT'22 **Poster Chair** and **Student Awards Chair**

SSDBM'21 **Session Chair**, *"Demo/Short Papers – DB/KB Applications"*

BDCAT'18 **Session Chair**, *"Big Data Infrastructures"*; *"Big Data Applications"*

FiCloud'18 **Program Track Chair**, *"Cloud Models"*

BDCAT'17 **Publicity Chair**

FiCloud'17 **Program Track Chair**, *"Energy Efficiency"*

Cloud'16 **Session Chairs**, *"Best Student Papers"*, *"CloudWare"*, and *"Cloud Data Storage"*

GFC'12 **General Chair**, GFC 2012 Workshop (Co-located with IEEE ICSG 2012)

CSC'12 **Session Chair**, *"Cloud Scheduling"*, IEEE CSC 2012

SSDBM'11 **Information Officer**

Technical Program Committees

BDCAT **PC Member**, 2026, 2024, 2023, 2021, 2020, 2019, 2018, 2016

CloudCom **PC Member**, 2012

HiPC **PC Member**, 2024, 2019, 2017, 2015

HPCC **PC Member**, 2010

IDEAS **PC Member**, 2026, 2024, 2022, 2021, 2020, 2019, 2018, 2017, 2016, 2015, 2014, 2012

ICCCN **PC Member**, 2016, 2015, 2014

ICPP	**PC Member**, 2015
IPDPS	**PC Member**, 2012
ISI	**PC Member**, 2013, 2012
ITiCSE	**PC Member**, 2015, 2014, 2013, 2012
ScienceCloud	**PC Member**, 2015, 2013, 2012
SIGCSE	**PC Member**, 2015, 2013, 2012

Journal Reviewer

COMP	**Reviewer**, Springer Journal of Computing
FGCS	**Reviewer**, Future Generation Computer Systems
IC	**Reviewer**, IEEE Internet Computing
ITPRO	**Reviewer**, IEEE IT Professional
JDPD	**Reviewer**, Journal on Distributed and Parallel Databases
JSS	**Reviewer**, Journal of Systems and Software
TPDS	**Reviewer**, IEEE Transactions on Parallel and Distributed Systems
TCC	**Reviewer**, IEEE Transactions on Cloud Computing
TKDE	**Reviewer**, IEEE Transactions on Knowledge and Data Engineering
XRDS	**Reviewer**, ACM Crossroads

Editorship

2012 - 2013	**Co-Editor**, IEEE Sustainable Computing Register
2009 - 2010	**Department Editor**, ACM Crossroads (XRDS)
2007 - 2009	**Associate Copy Editor**, ACM Crossroads (XRDS)

Community Service

2011 - 2017	**Faculty Mentor**, Google Summer of Code (GSoC)
2013	**Judge**, Portland Public Schools Science Expo
2012	**Faculty Lead**, MESA Engineering Day
2011	**Broader Engagement Mentor**, Supercomputing Conference (SC'11)
2011	**Judge**, Intel North West Science Expo
2011	**Lead Judge**, MESA Day
2011	**Faculty Lead**, MESA Engineering Day
2010	**Broader Engagement Mentor**, Supercomputing Conference (SC'10)
2010	**Faculty Lead**, MESA Engineering Day
2008 - 2009	**Teaching Orientation Facilitator**, The Ohio State University

Research Student Advising

Research Alumni and First Employment (If Known)

2026 **Lily Gustafson (BS'27)**,
"From Adverse Events to Lived Experience: A Comparative Analysis of Patient and Institutional Reporting for Psychiatric Medications." (With Marissa Masden)

2025 **Alex Kirner (BS'26)**,
"Support and Acceleration of Join Queries over Bitmap Indices"

2024 **Julia Kaeppel (BS'24)**,
"Workload-Aware Cache Management of Bitmap Indices"

2022 **Colin Monaghan (BS'22)**, Flock Freight
"New Replacement Policies for Semantic Database Caching"

2021 **Manya Mutschler-Aldine (BS'21)**, Marigold
"Implementation and Evaluation of MIN and CLOCK for Bitmap Caching"

2020 **Sarah Walling-Bell (BS'20)**, Center for Neurotechnology @ UW
"EEG Experiment Scripting Tool for Novice Programmers"

2020 **Sarah McClain (BS'20)**, Panorama Education, Boston
"Toward Semantic Caching of Bitmap Indices"

2019 **Jared Polonitza (BS'19)**, Chalice Network
"Toward a Transactive Energy Future"

2018 **Sam Burdick (BS'18)**, Amazon
"Fault-Tolerant Query Execution over Distributed Bitmap Indices"

2018 **Jahrme Risner (BS'18)**, Google
"Fault-Tolerant Query Execution over Distributed Bitmap Indices"

2018 **Patrick Ryan (BS'18)**, T-Mobile, Bellevue
"Optimizing Byte-Aligned Bitmap Code"

2018 **Jack Burns (BS'18)**, Loopie
"An Optimal Bitmap Binning Method for Resolving Aggregate Queries"

2018 **Jiawen (Christine) Li (BS'18)**, Student at Columbia University
"Modeling Power Usage of Bitmap Processing Algorithms"

2017 **Chili Johnson (BS'17)**
"Hadoop Migration for Energy Conservation"

2017 **Rachel Hirsch (BS'17)**, Student at Colorado State University
"Live Data Compression, Caching, and Querying of Bitmap Indices"

2016 **Alexia Ingerson (BS'16)**, Intel, Hillsboro
"Analyzing Caching Patterns for Bitmap Processing"

2016 **Brandon Roberts (BS'16)**, Milliman – MedInsight
"Mobile Applications for ICD9/10 Conversions"

2015 **Ben McCamish (MS'15)**, Clinical Professor at Washington State University
"Managing PMU Data Sets with Bitmap Indexes"

2014 **Bo Wang (MS'14)**, Microsoft, Redmond
"Design & Implementation of an Energy-Aware App for Scientific Field Studies"

2014 **Eriko Otsuka (MS'14)**, Clark College
"Design and Evaluation of a Twitter Hashtag Recommendation System"

2014 **Gabriel Dunham (BS'14)**, Plexsys
"Modeling Parallel Molecular Simulations on Amazon EC2"

2014 **Sean Lim (BS'14)**, Dell EMC, Seattle
"SkySurvey Data Collection and Analysis"

2014 **Helena Lucia (BS'14)**, Intel, Hillsboro
"An Android Mobile App for Barcode-Based Search of Food Allergies"

2013 **Fredton Doan (MS'13)**, Clark County Washington
"Workload-Driven Bitmap Recompression for Real-Time Query Acceleration"

2012 **Farhana Kabir (MS'12)**, Intel, Hillsboro
"Reconciling Cost and Performance Objectives for Elastic Web Caches"

2012 **Travis Hall (MS'12)**, VMware, Seattle
"A Cost-Driven Replacement Policy for a Hierarchical Key-Value Store"

2012 **Skylar Hiebert (BS'12)**, WebTrends, Portland
"FreeQuiz"

Other Students (As Thesis Committee Member)

2016 Xiaolei Huang (MS'16, PhD'21, University of Colorado at Boulder)

2014 Miles Histand (BS'14, Portland State University)

2014 Tekin Biçer (PhD'14, The Ohio State University)

2014 Huy Tran (MS'14, WSU Vancouver)

2014 Hoang Le (MS'14, WSU Vancouver)

2013 Evan Dickinson (MS'13, WSU Vancouver)

2013 Sanchit Aggarwal (MS'13, University of Pennsylvania)

2011 Matt Henry (MS'11, WSU Vancouver)

2011 Fabian Corrales (BS'11, University of Puget Sound)

2010 Gang Lu (MS'10, WSU Vancouver)

2010 Apeksha Shetty (MS'10, The Ohio State University)

Awards and Honors

For Teaching and Mentoring

2024 **Influential Faculty of Color Award**, University of Puget Sound

2022 **President's Award for Excellence in Teaching**, University of Puget Sound

2010 **Diversity Council Award**, WSU Vancouver

2010 **Eleanor Quinlan Memorial Award for Excellence in Teaching**,
Computer Science and Engineering, The Ohio State University

2009 **Preparing Future Faculty (PFF) Fellow**, Ohio State and Denison University

For Scholarship

2026 **John Lantz Senior Fellowship for Research**, University of Puget Sound

2023 **Best Paper Award**, IEEE/ACM BDCAT 2023

2020 **Elevation to Senior Member Grade**, IEEE

2019 **Best Paper Finalist**, IEEE/ACM BDCAT 2019

2018 **Best Paper Finalist**, IEEE/ACM BDCAT 2018

2018 **Best Poster Runner-Up** (with Jared Polonitza), CCSC-NW 2018

2017 **Bartanen Research Award**, University of Puget Sound

2015 **Best Poster Award** (with Alexia Ingerson), CCSC-NW 2015

2009 **Upsilon Pi Epsilon**, Honor Society for the Computing & Information Disciplines

2004 **Best Student Paper Award**, ITE 2004

2004 **Best Student Paper Award**, IEEE ITCC 2004

Professional Affiliations

ACM **Member**, Association for Computing Machinery

SIGCSE **Member**, ACM Special Interest Group on Computer Science Education

SIGHPC **Member**, ACM Special Interest Group on High-Performance Computing

SIGMOD **Member**, ACM Special Interest Group on Management of Data

IEEE **Senior Member**, Institute of Electrical and Electronics Engineers